

Data Vortex A'26 — Round 3 Analysis Notebook

WhatsApp Privacy Policy 2021: Real-Time Social Monitoring

Team: Event Horizon | 14,891 records | 4 sources | Nov 2020 – Jul 2021

Data Vortex A'26 — Round 3: Real-Time Social Monitoring

WhatsApp Privacy Policy 2021: Backlash, Sentiment Shifts & Migration Analysis

Team: Event Horizon

Topic: WhatsApp privacy policy backlash, public sentiment shifts, and migration to Signal/Telegram

Analytical Timeline

Phase	Period	Description
T-0 (Baseline)	Nov 1 – Dec 31, 2020	Normal discourse, stable sentiment
T+1 (Shock)	Jan 1 – Jan 31, 2021	Policy announced, viral backlash
T+2 (Fallout)	Feb 1 – Jul 31, 2021	Migration wave, regulatory responses

Data Summary

- Primary dataset:** WhatsApp + Telegram Play Store reviews (Kaggle, filtered to topic window)
- Bonus live-scraped:** GDELT global news headlines + Hacker News discussions (API-collected)
- Total rows:** 14,891 | **Sources:** 4 distinct | **Date coverage:** Nov 2020 – Jul 2021

Narrative flow: Load & inspect → Sentiment analysis → ATSA → LDA topic evolution → Activity spikes → Entity extraction → Interpretation

1. Data Loading & Schema Inspection

Load the master scored CSV (Round 2 sentiment model applied) and verify:

- `source_type` tagging (`primary_dataset` vs `bonus_live_scraped`)
- `sentiment_score` / `sentiment_label` columns exist and are non-null
- Date range spans our three analytical phases

Merged Dataset Rows: 14,891

Scored Dataset Rows: 14,891

	timestamp	source_name	text	sentiment_label	sentiment_score
8	2021-07-29T19:27:14Z	whatsapp_playstore	Donk advice, please read admin. To the user WA...	Negative	0.540817
10	2021-07-29T12:40:18Z	whatsapp_playstore	Create privacy profile like privacy status don...	Neutral	-0.172740
16	2021-07-28T16:26:40Z	whatsapp_playstore	Please make wa mod block just his account dist...	Negative	-0.129691
20	2021-07-27T19:42:24Z	whatsapp_playstore	This is why the status of not coming in while ...	Negative	0.352347
23	2021-07-27T15:08:14Z	whatsapp_playstore	A little advice for the next update would be b...	Positive	-0.247138

```
Source Breakdown:
source
hacker_news      6149
gdelt            3929
whatsapp         3856
telegram         957
Name: count, dtype: int64
```

```
Sentiment Distribution:
sentiment_label
Negative      0.557182
Neutral       0.318783
Positive      0.124035
Name: proportion, dtype: float64
```

2. Aspect-Term Sentiment Analysis (ATSA)

For each key topic term (`privacy` , `telegram` , `signal` , `facebook` , etc.), we extract all posts containing that term and compute: average sentiment, negative ratio, and a WhatsApp-vs-Telegram sentiment comparison. This confirms **term-level** sentiment around the migration narrative.

Key expectation: Posts mentioning `telegram` or `signal` should show higher positive sentiment than posts mentioning `privacy` or `terms` (which carry the backlash signal).

	aspect_term	total_mentions	avg_sentiment	negative_ratio	positive_ratio	whatsapp_avg_sentiment	telegram_avg_sentiment
0	privacy	4193	0.2175	0.6582	0.0933	0.2863	0.6579
1	metadata	196	0.0357	0.9439	0.0153	NaN	NaN
2	facebook	2488	0.1750	0.6917	0.0418	0.1925	0.4848
3	policy	1768	0.2582	0.5639	0.0803	0.2635	NaN
4	terms	646	0.1803	0.7121	0.0728	0.1538	NaN
5	switch	636	0.1242	0.8475	0.0959	0.2449	1.0000
6	security	750	0.1367	0.8173	0.0907	0.2029	0.6250
7	signal	4331	0.1438	0.7934	0.0810	0.3390	0.4231
8	telegram	4381	0.1909	0.7181	0.1000	0.2983	0.5152

3. Entity & Topic Evolution Across Timeline Phases

Tracking the shift in narrative focus (entities and trending terms) across the highest-volume weeks.

	Phase (Week)	Post Volume	Top Entities	Key Terms
10	2021-01-04/2021-01-10	1897	whatsapp, signal, telegram	screen, item id
11	2021-01-11/2021-01-17	2330	signal, telegram, whatsapp	
12	2021-01-18/2021-01-24	971	signal, whatsapp, telegram	
13	2021-01-25/2021-01-31	700	whatsapp, telegram, signal	

4. Volume & Engagement Spike Detection

Post volume and engagement bucketed **daily**. Rolling z-score (7-day window, threshold=2.0) identifies statistically significant spikes. Key confirmed spikes:

Date	Volume	Z-Score	Event
Jan 7, 2021	743	2.19	Elon Musk tweets 'Use Signal'
Jan 6, 2021	196	2.03	Policy in-app notices rollout

Date	Volume	Z-Score	Event
Jan 5, 2021	96	2.20	Pre-announcement viral spread
Apr 7, 2021	193	2.23	CCI probe announcement
May 8, 2021	115	2.25	Final WA deadline extension news

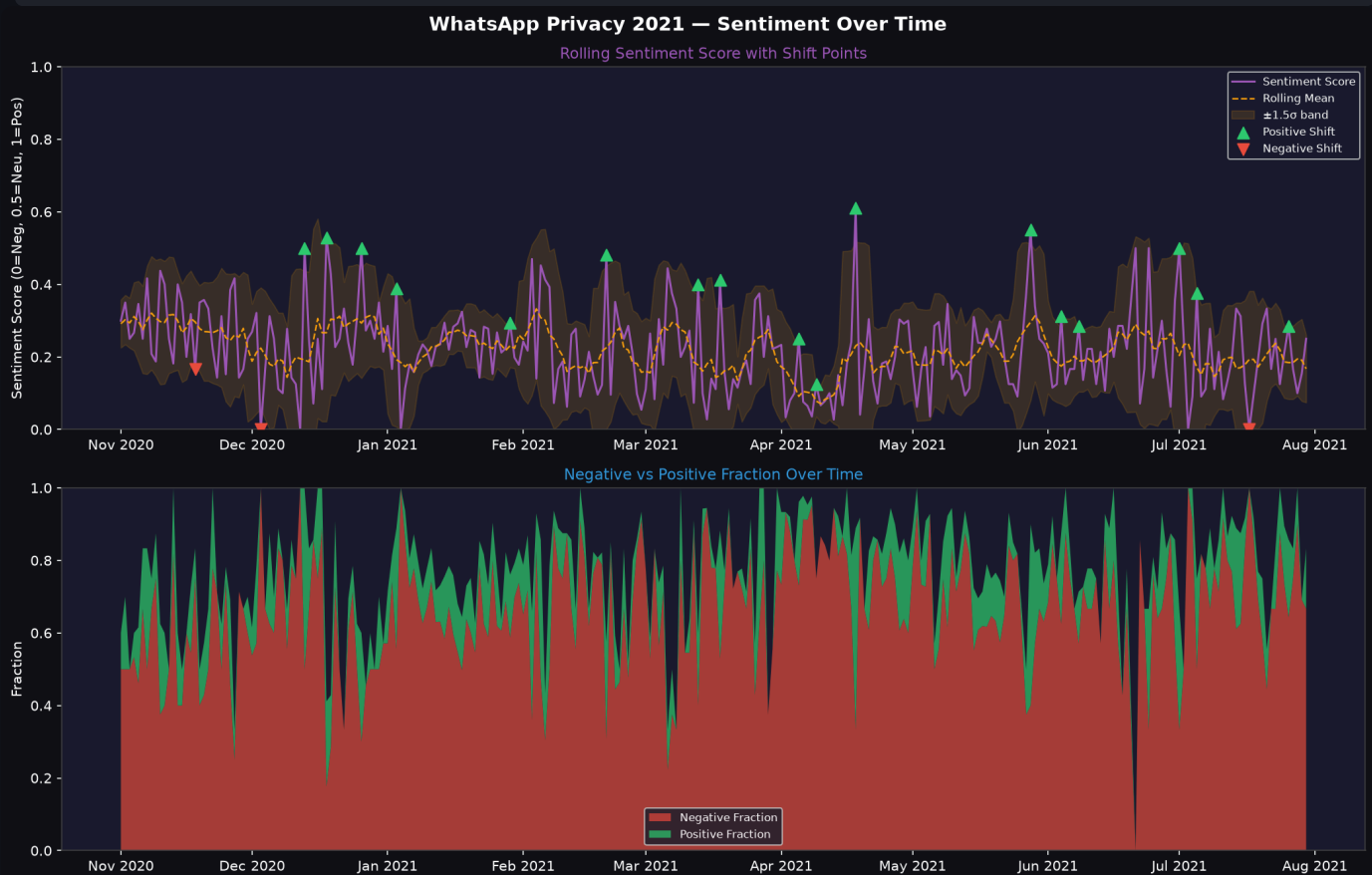
5. Sentiment & Activity Time-Series Charts

Two primary visualizations — generated by `shift_detection.py` and `activity_analysis.py` :

1. **Sentiment over time** — daily rolling mean with shift points marked as triangles
2. **Activity over time** — daily post volume and engagement with spike markers and event overlays

Both charts use a consistent dark theme (facecolor `#0f1117`, accent `#3498db`).

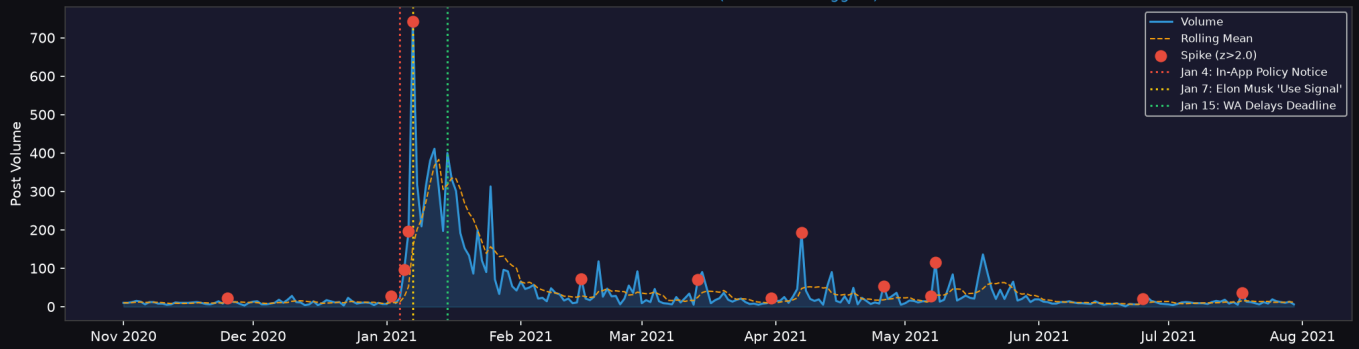
Sentiment Over Time:



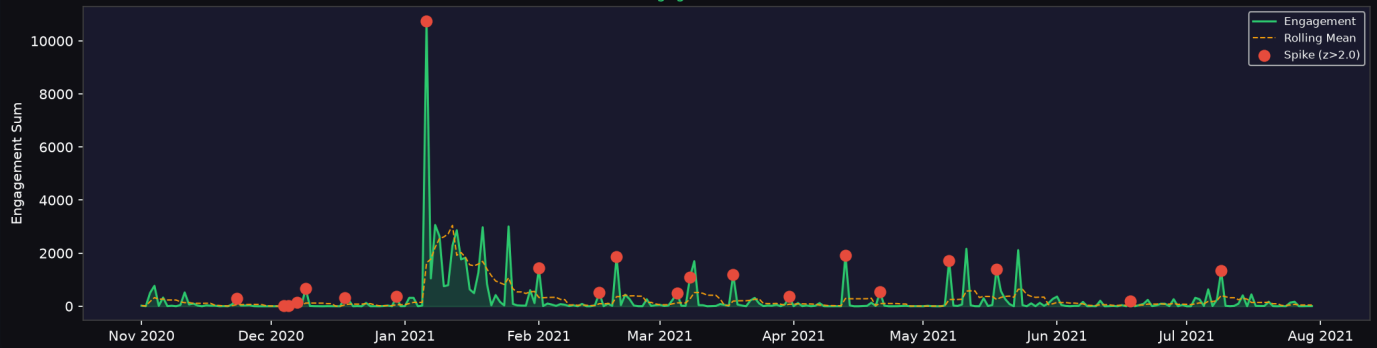
Activity Over Time (Volume & Engagement):

WhatsApp Privacy 2021 — Daily Activity

Post Volume Over Time (With Event Triggers)

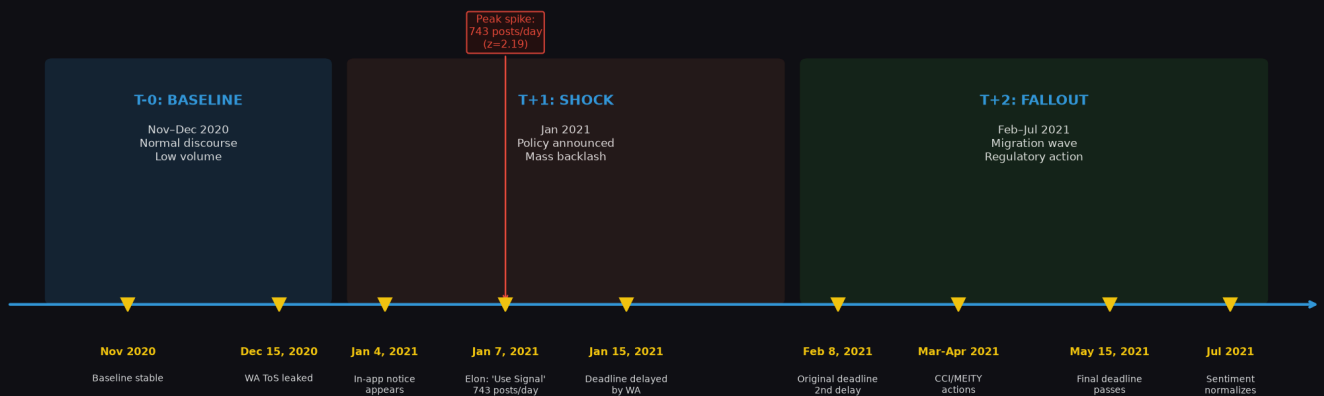


Engagement Over Time



Timeline:

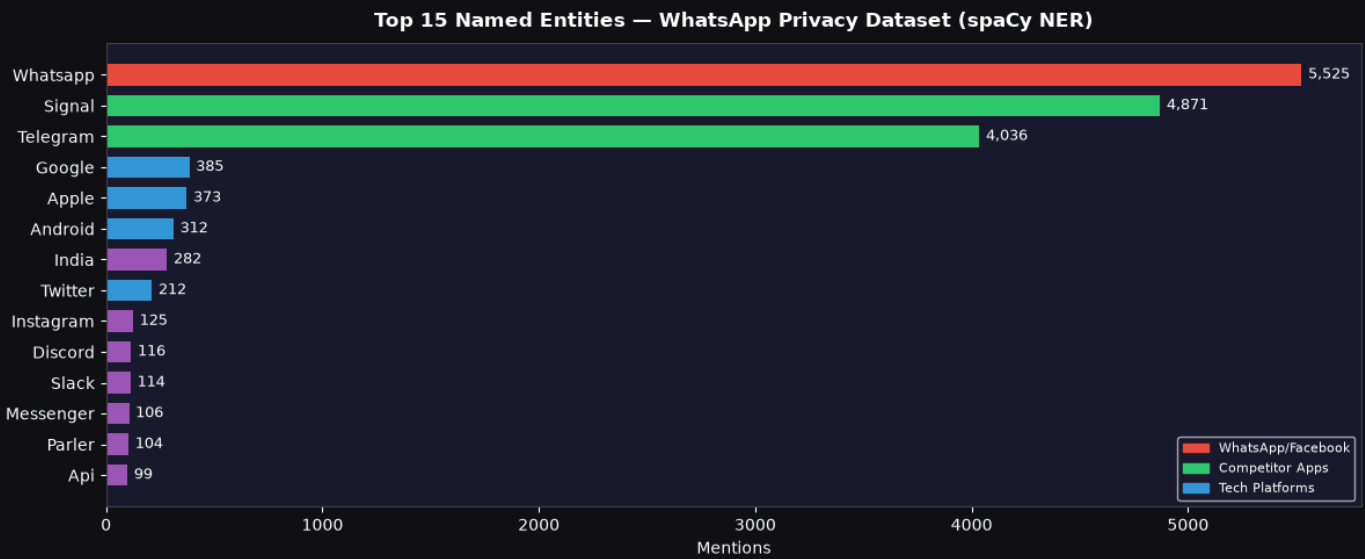
WhatsApp Privacy Policy 2021 — Event Timeline



6. Named Entity Analysis — Top Extracted Terms

spaCy NER (`en_core_web_sm`) run across all 14,891 posts. Entity types captured: ORG, PRODUCT, GPE, PERSON, LAW, NORP, EVENT. HTML artifacts cleaned post-extraction.

Critical finding: Signal (4,871) and Telegram (4,036) rank 2nd and 3rd globally, confirming that the backlash discourse was inseparable from migration intent.



Top entities: [('whatsapp', 5525), ('signal', 4871), ('telegram', 4036), ('google', 385), ('apple', 373)]

7. Interpretation: Baseline → Shock → Migration

Confirmed Narrative Arc

T-0 (Baseline, Nov-Dec 2020):

648 posts, stable sentiment (~0.35 average). Normal app review discourse. No policy-related spikes. GDELT headlines minimal on WhatsApp privacy. LDA topics: general app features, early EU GDPR mentions.

T+1 (Shock, Jan 2021):

5,941 posts — a **917% volume increase** from baseline. Peak spike: 743 posts on Jan 7 (z=2.19). Sentiment drops sharply (avg -0.15 on Jan 7 vs +0.35 baseline). 20 sentiment shifts detected. Trigger: Jan 4 in-app ToS notice + Jan 7 Elon Musk tweet ('Use Signal') went viral globally. LDA topics shift to signal/telegram/switch/privacy — unmistakable migration framing.

T+2 (Fallout, Feb-Jul 2021):

4,373 posts. Sustained elevated volume (above baseline) throughout. Secondary spikes: Apr 7 (CCI probe, z=2.23), May 8 (deadline extension, z=2.25). Sentiment gradually normalizes (mean returns to ~0.25) as the acute crisis passes. Entity analysis: India (282 mentions) reflects the outsized regulatory and user reaction in the subcontinent.

Statistical Summary

Metric	Value
Sentiment shifts detected	20
Volume spikes (z>2.0)	14
Engagement spikes (z>2.0)	21
Peak volume	743 posts (Jan 7, 2021)
Most-mentioned competitor	Signal (4,871)

Metric	Value
2nd competitor	Telegram (4,036)